

定積分と微分

a を定数とすると

$$\frac{d}{dx} \int_a^x f(t) dt = f(x), \quad \frac{d}{dx} \int_x^a f(t) dt = -f(x)$$

(例1) 次の関数を x について微分せよ。

(1) $y = \int_0^x t^3 e^t dt$

$$y' = \frac{d}{dx} \int_0^x t^3 e^t dt \\ = x^3 e^x$$

(2) $y = \int_x^1 e^t \sin t dt$

$$y' = \frac{d}{dx} \int_x^1 e^t \sin t dt \\ = -e^x \sin x \quad \leftarrow e^x \sin x \text{ とはい!}$$

(3) $y = \int_{\pi}^x (x+t) \cos t dt$

$$y = x \int_{\pi}^x \cos t dt + \int_{\pi}^x t \cos t dt \quad \leftarrow t \text{ と無関係な文字 } x \text{ は定数とみなせる}$$

よ

$$y' = (x) \int_{\pi}^x \cos t dt + x \left(\frac{d}{dx} \int_{\pi}^x \cos t dt \right) + \frac{d}{dx} \int_{\pi}^x t \cos t dt \\ = \int_{\pi}^x \cos t dt + x \cos x + x \cos x \\ = [\sin t]_{\pi}^x + 2x \cos x \\ = \sin x + 2x \cos x$$

(例2) 次の関数を x について微分せよ。

(1) $y = \int_x^{2x} \sin^2 t dt$

$$F(t) = \int \sin^2 t dt \text{ とおくと } F'(t) = \sin^2 t \text{ であるから}$$

$$y = F(2x) - F(x)$$

よ

$$y' = \frac{d}{dx} \{ F(2x) - F(x) \} \\ = 2F'(2x) - F'(x) \\ = 2\sin^2 2x - \sin^2 x$$

(2) $y = \int_x^{x^2} (x+t) \log t dt \quad (x > 0)$

$$y = x \int_x^{x^2} \log t dt + \int_x^{x^2} t \log t dt$$

よ

$$y' = (x) \int_x^{x^2} \log t dt + x \left(\frac{d}{dx} \int_x^{x^2} \log t dt \right) + \frac{d}{dx} \int_x^{x^2} t \log t dt$$

そこで

$$\int_x^{x^2} t \log t dt = [t \log t - \frac{1}{2} t^2]_x^{x^2} \\ = (x^2 \log x^2 - \frac{1}{2} x^2) - (x \log x - \frac{1}{2} x^2) \\ = (2x^2 - x) \log x - \frac{1}{2} x^2 + \frac{1}{2} x^2$$

$$\frac{d}{dx} \int_x^{x^2} t \log t dt = \{ (2x^2 - x) \log x - \frac{1}{2} x^2 + \frac{1}{2} x^2 \}' \\ = (4x - 1) \log x$$

また $F(t) = \int t \log t dt$ とおくと $F'(t) = t \log t$ であるから

$$\int_x^{x^2} t \log t dt = F(x^2) - F(x)$$

よ

$$\frac{d}{dx} \int_x^{x^2} t \log t dt = \frac{d}{dx} \{ F(x^2) - F(x) \} \\ = 2x F'(x^2) - F'(x) \\ = 2x \cdot x^2 \log x^2 - x \log x \\ = (4x^3 - x) \log x$$

以上よ

$$y' = \{ (2x^2 - x) \log x - \frac{1}{2} x^2 + \frac{1}{2} x^2 \} + x \{ (4x - 1) \log x \} + (4x^3 - x) \log x \\ = (4x^3 + 6x^2 - 3x) \log x - \frac{1}{2} x^2 + \frac{1}{2} x^2$$